

BITS PILANI K.K. BIRLA GOA CAMPUS

Department of Physics

What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 3: Canonical Equations

Group-B

Priyansh Rawat	2023B5A70968G
Mayank Sinha	2023B5A70942G
Samarth Bhatnagar	2023B5A70853G

Under Professor: Rudranil Basu

29 October 2026

1 Legendre transform with derivative mixing

For one real Fourier sector,

$$L_k = \frac{a^2}{2}(q'^2 + \sigma'^2) - \frac{a^2k^2}{2}(q^2 + \sigma^2) - \frac{a^4m^2}{2}\sigma^2 + a^3\rho q'\sigma. \tag{1}$$

Differentiating the Lagrangian, rather than identifying momenta from the unmixed theory, gives

$$\pi_q = a^2q' + a^3\rho\sigma, \qquad \pi_\sigma = a^2\sigma'. \tag{2}$$

Hence $q' = (\pi_q - a^3\rho\sigma)/a^2$ and $\sigma' = \pi_\sigma/a^2$. Substituting these velocities into $H_k = \pi_q q' + \pi_\sigma\sigma' - L_k$ yields

$$\boxed{H_k = \frac{(\pi_q - a^3\rho\sigma)^2 + \pi_\sigma^2}{2a^2} + \frac{a^2k^2}{2}(q^2 + \sigma^2) + \frac{a^4m^2}{2}\sigma^2.} \tag{3}$$

The square contains $a^4\rho^2\sigma^2/2$. Dropping it changes the dynamics even though the mixing in L_k is linear in ρ. A symbolic calculation independently expands the Legendre transform and verifies exact equality with this Hamiltonian. Hamilton's equations are

$$q' = (\pi_q - a^3\rho\sigma)/a^2, \qquad \pi_q' = -a^2k^2q, \tag{4}$$

$$\sigma' = \pi_\sigma/a^2, \qquad \pi_\sigma' = a\rho\pi_q - a^2k^2\sigma - a^4(m^2 + \rho^2)\sigma. \tag{5}$$

They also fix the initial velocity of a spectator-only column: an initially vanishing π_q implies a nonzero q' when $\sigma \neq 0$.

2 Dimensionless equations and an independent formulation

Set $x = -k\eta$, $M = m/H$, $g = \rho/H$, and write the two independent mode columns as $q_k^a = Hk^{-3/2}Q_a$, $\sigma_k^a = Hk^{-3/2}\Sigma_a$. The canonical equations are

$$\partial_x Q = g\Sigma/x - x^2\Pi_q, \qquad \partial_x \Pi_q = Q/x^2, \tag{6}$$

$$\partial_x \Sigma = -x^2\Pi_\sigma, \qquad \partial_x \Pi_\sigma = \Sigma/x^2 + (M^2 + g^2)\Sigma/x^4 - g\Pi_q/x. \tag{7}$$

For $N = -\ln x$, $\partial_N = -x\partial_x$. Eliminating momenta gives the independently implemented velocity equations

$$\ddot{Q} + 3\dot{Q} + x^2 Q = -g(\dot{\Sigma} + 3\Sigma), \tag{8}$$

$$\ddot{\Sigma} + 3\dot{\Sigma} + (x^2 + M^2)\Sigma = g\dot{Q}, \tag{9}$$

$$\Pi_q = (\dot{Q} + g\Sigma)/x^3, \qquad \Pi_\sigma = \dot{\Sigma}/x^3. \tag{10}$$

Both implementations evolve two complex columns, preserving the phases needed later. At $M = 2$, $g = 0.1$, $X = 100$, $x_f = 10^{-3}$, DOP853 with relative/absolute tolerances $10^{-10}/10^{-12}$ gives a maximum scaled column difference of 2.010×10^{-7} and an entropy difference of 5.111×10^{-7}. The scaled difference is $|u_N - u_x|/(1 + |u_N|)$; it remains meaningful near a zero of a mode column. Production precision and initial-state tests are established in the next steps.

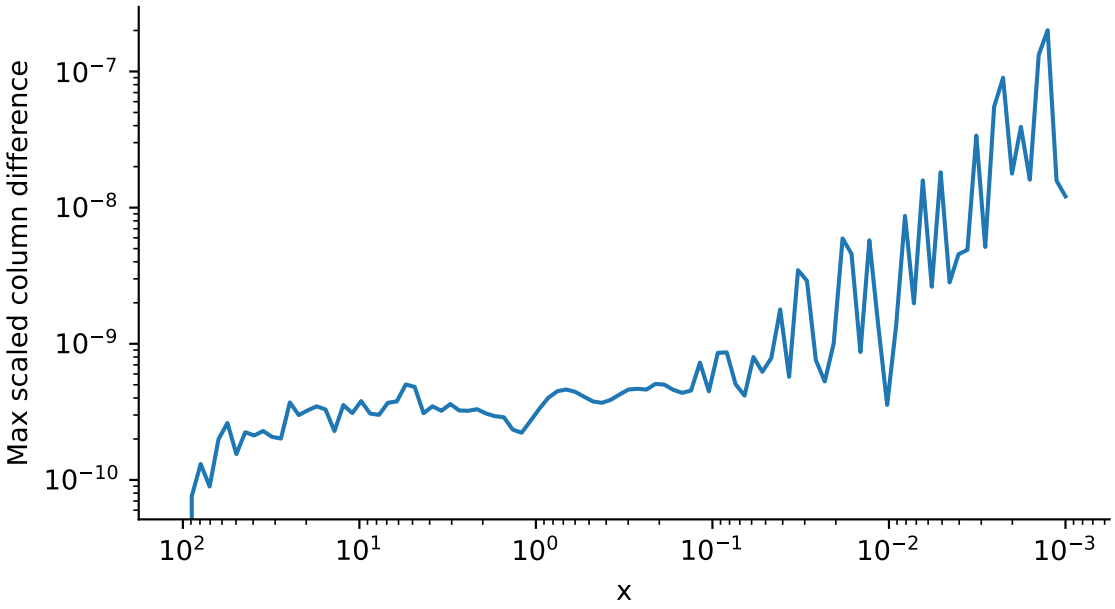


Figure 1: Agreement of canonical and velocity evolutions on identical finite-time initial data. The comparison is made before any covariance normalisation.

Code and supporting data: step3 on GitHub.

Contributions (equal)

Member	Contribution
Priyansh Rawat	Legendre transform and Hamiltonian derivation.
Samarth Bhatnagar	Dimensionless and velocity equations; convention checks.
Mayank Sinha	Independent numerical formulations and agreement tests.